自动驾驶控制算法第五讲

$$\begin{pmatrix} \dot{e_d} \\ \ddot{e_d} \\ \dot{e_\varphi} \\ \ddot{e_\varphi} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & \frac{C_{\alpha f}+C_{\alpha r}}{mv_x} & -\frac{C_{\alpha f}+C_{\alpha r}}{m} & \frac{aC_{\alpha f}-bC_{\alpha r}}{mv_x} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{aC_{\alpha f}-bC_{\alpha r}}{I_z v_x} & -\frac{aC_{\alpha f}-bC_{\alpha r}}{I_z} & \frac{a^2C_{\alpha f}+b^2C_{\alpha r}}{I_z v_x} \end{pmatrix} \begin{pmatrix} e_d \\ \dot{e_d} \\ e_\varphi \\ \dot{e_\varphi} \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{C_{\alpha f}}{m} \\ 0 \\ -\frac{aC_{\alpha f}}{I_z} \end{pmatrix} \delta + \begin{pmatrix} 0 \\ \frac{aC_{\alpha f}-bC_{\alpha r}}{mv_x} - v_x \\ 0 \\ \frac{a^2C_{\alpha f}+b^2C_{\alpha r}}{I_z v_x} \end{pmatrix} \dot{\theta}_r$$

$\dot{e_{rr}} = Ae_{rr} + Bu + C\dot{\theta}_r$ 暂时先忽略 $C\dot{\theta}_r$

$\dot{e_{rr}} = Ae_{rr} + Bu$

matlab `lqr(A,B,Q,R)` `dlqr(A,B,Q,R)` 离散 lqr discrete

dlqr 原理 (连续 lqr 原理涉及泛函与变分法，这在力学系中也是相当高级的知识)

dlqr 原理只涉及拉格朗日乘子法 求 $f(x,y)$ 在约束 $g(x,y)=0$ 条件下的极值

$L(x,y) = f(x,y) + \lambda g(x,y)$

$\frac{\partial L}{\partial x}=0 \quad \frac{\partial L}{\partial y}=0 \quad \frac{\partial L}{\partial \lambda}=0$ (高等数学)

dlqr 问题 $\dot{x} = Ax + Bu \xrightarrow{\text{discrete}} x_{k+1} = \bar{A}x_k + \bar{B}u_k$

$J = \sum_{k=0}^{+\infty}(x_k^T Q x_k + u_k^T R u_k)$ 在约束 $x_{k+1} = \bar{A}x_k + \bar{B}u_k$ 下的极小值

问题一： $\dot{x} = Ax + Bu \Rightarrow x_{k+1} = \bar{A}x_k + \bar{B}u_k$

问题二： 求解使得 J 在约束 $x_{k+1} = \bar{A}x_k + \bar{B}u_k$ 下取极小值的 u_k

离散化：向前欧拉法，中点欧拉法 apollo 是两者的混合

$\dot{x} = Ax$ 积分 $\int_t^{t+dt} \dot{x}\,dt = \int_t^{t+dt} Ax\,dt \Rightarrow x(t+dt) - x(t) = A x(\xi)\,dt \quad \xi \in (t, t+dt)$

积分中值定理

$X(t+dt) = X(t) + A X(\xi) dt \quad \xi \in (t, t+dt)$ 精确离散化

向前欧拉法 $X(\xi) = X(t) \Rightarrow X(t+dt) = (I + Adt) X(t)$

向后欧拉法 $X(\xi) = X(t+dt) \Rightarrow X(t+dt) = (I - Adt)^{-1} X(t)$

中点欧拉法 $X(\xi) = \frac{X(t) + X(t+dt)}{2} \Rightarrow X(t+dt) = (I - \frac{Adt}{2})^{-1} (I + \frac{Adt}{2}) X(t)$

$\dot{X} = Ax + Bu \xrightarrow{\text{积分}} \int_t^{t+dt} \dot{x} = \int_t^{t+dt} (Ax + Bu) dt \Rightarrow X(t+dt) - X(t) = A X(\xi) dt + Bu(\xi) dt$

$X(t+dt) = A X(\xi) dt + X(t) + B u(\xi) dt$

中点欧拉法　　向前欧拉法，因为 $u(t+dt)$ 不知道

$= Adt \left(\frac{X(t+dt) + X(t)}{2} \right) + X(t) + Bdt\, u(t)$

$(I - \frac{Adt}{2}) X(t+dt) = (I + \frac{Adt}{2}) X(t) + Bdt\, u(t)$

$X(t+dt) = (I - \frac{Adt}{2})^{-1} (I + \frac{Adt}{2}) X(t) + (I - \frac{Adt}{2})^{-1} Bdt\, u(t)$

$\approx (I - \frac{Adt}{2})^{-1} (I + \frac{Adt}{2}) X(t) + Bdt\, u(t)$

$X(t+dt) = (I - \frac{Adt}{2})^{-1} (I + \frac{Adt}{2}) X(t) + Bdt\, u(t)$

$dt = 0.01 \quad X(k+1) = \bar{A} X(k) + \bar{B} u(k)$

$\bar{A} = (I - \frac{Adt}{2})^{-1} (I + \frac{Adt}{2}) \quad \bar{B} = Bdt \quad \dot{x} = Ax + Bu \Rightarrow x_{k+1} = \bar{A} x_k + \bar{B} u_k$

dlqr 的解法

$J = \sum_{k=0}^{\infty} (x_k^T Q x_k + u_k^T R u_k) \quad$ 约束 $x_{k+1} = A x_k + B u_k$

$J = \sum_{k=0}^{n-1} (x_k^T Q x_k + u_k^T R u_k) + x_n^T Q x_n$

约束

$$\left.\begin{aligned} & Ax_0 + Bu_0 - x_1 = 0 \\ & Ax_1 + Bu_1 - x_2 = 0 \\ & \vdots \\ & Ax_{n-1} + Bu_{n-1} - x_n = 0 \end{aligned}\right\} \text{覆盖 } x_0 \sim x_n \quad u_0 \sim u_{n-1}$$

$$J = \sum_{k=0}^{n-1} (x_k^T Q x_k + u_k^T R u_k) + x_n^T Q x_n + \sum_{k=0}^{n-1} \lambda_{k+1}^T (A x_k + B u_k - x_{k+1})$$

$$= \sum_{k=0}^{n-1} \underbrace{\left[x_k^T Q x_k + u_k^T R u_k + \lambda_{k+1}^T (A x_k + B u_k) \right]}_{H_k} - \lambda_{k+1}^T x_{k+1} + x_n^T Q x_n$$

设 $H_k = x_k^T Q x_k + u_k^T R u_k + \lambda_{k+1}^T (A x_k + B u_k)$

$$J = \sum_{k=0}^{n-1} (H_k - \lambda_{k+1}^T x_{k+1}) + x_n^T Q x_n$$

$$= \underbrace{H_0 + H_1 + \cdots + H_{n-1}}_{0 \sim n-1} + \underbrace{(-\lambda_1^T x_1 - \lambda_2^T x_2 - \cdots}_{1 \sim n-1} \underbrace{\lambda_n^T x_n}_{n}) + x_n^T Q x_n$$

$$= \sum_{k=0}^{n-1} H_k + \underbrace{\sum_{k=1}^{n-1} (-\lambda_k^T x_k)} + (-\lambda_n^T x_n) + x_n^T Q x_n + \underbrace{(-\lambda_0^T x_0)} - (-\lambda_0^T x_0)$$

$$= \sum_{k=0}^{n-1} H_k + \sum_{k=0}^{n-1} (-\lambda_k^T x_k) - \lambda_n^T x_n + x_n^T Q x_n + \lambda_0^T x_0$$

$$= \sum_{k=0}^{n-1} (H_k - \lambda_k^T x_k) + x_n^T Q x_n - \lambda_n^T x_n + \lambda_0^T x_0$$

向量导数

$$\frac{\partial (x^T A)}{\partial x} = A \qquad \frac{\partial (Ax)}{\partial x} = A^T \qquad \frac{\partial (x^T A x)}{\partial x} = (A + A^T)x \ (\text{若 } A \text{ 为对称矩阵}) = 2Ax$$

$$\frac{\partial J}{\partial x_0} = \vec{0} \qquad \frac{\partial \lambda_0^T x_0}{\partial x_0} = \vec{0} \Rightarrow \lambda_0 = \vec{0}$$

$$\frac{\partial J}{\partial x_k} = \vec{0} \Rightarrow \lambda_k = \frac{\partial H_k}{\partial x_k} \quad (k = 1, 2, 3 \cdots n-1)$$

$$\frac{\partial J}{\partial u_k} = \vec{0} \Rightarrow \frac{\partial H_k}{\partial u_k} = \vec{0} \quad (k = 1, 2, 3 \cdots n-1)$$

$$\frac{\partial J}{\partial x_n} = \vec{0} \Rightarrow \frac{\partial (x_n^T Q x_n - \lambda_n^T x_n)}{\partial x_n} = 0 \qquad 2Q x_n - \lambda_n = 0$$

$$\frac{\partial J}{\partial \lambda_k} = 0 \Rightarrow x_{k+1} = A x_k + B u_k \quad (k = 1, 2, 3 \cdots n-1)$$

$$H_k = x_k^T Q x_k + u_k^T R u_k + \lambda_{k+1}^T (A x_k + B u_k)$$

$$\frac{\partial H_k}{\partial x_k} = 2Q x_k + A^T \lambda_{k+1}$$

$$\frac{\partial H_k}{\partial u_k} = 2R u_k + B^T \lambda_{k+1}$$

$$\begin{cases} \lambda_k = 2Qx_k + A^T\lambda_{k+1} & ① \quad (k=1,2,\cdots n-1) \\ u_k = -\frac{1}{2}R^{-1}B^T\lambda_{k+1} & ② \quad (k=1,2,\cdots n-1) \\ x_{k+1} = Ax_k + Bu_k & ③ \quad (k=1,2,\cdots n-1) \\ \lambda_n = 2Qx_n & ④ \end{cases}$$

递推式

④带入②的($k=n-1$式) $u_{n-1} = -\frac{1}{2}R^{-1}B^T(2Qx_n) = -R^{-1}B^TQx_n$ ⑤

⑤带入③的$k=n-1$式 $x_n = Ax_{n-1} + Bu_{n-1}$

$= Ax_{n-1} + (-BR^{-1}B^TQx_n)$

$x_n = (I + BR^{-1}B^TQ)^{-1}Ax_{n-1}$ I：单位矩阵 ⑥

④带入①的$k=n-1$式 $\lambda_{n-1} = 2Qx_{n-1} + A^T\lambda_n$

$= 2Qx_{n-1} + A^T2Qx_n$ ⑦

⑥带入⑦ $\lambda_{n-1} = 2Qx_{n-1} + 2A^TQ(I + BR^{-1}B^TQ)^{-1}Ax_{n-1}$

$= 2[Q + A^TQ(I + BR^{-1}B^TQ)^{-1}A]x_{n-1}$ ⑧

④与⑧比较 $\lambda_n = 2Qx_n$ $\lambda_{n-1} = 2(\underbrace{\quad\quad}_{Q_{n-1}})x_{n-1}$

设 $\lambda_k = 2P_kx_k$ $(k=1,2,3,\cdots n)$ 则由④

$P_n = Q$

④→② $u_{n-1} = -R^{-1}B^TP_nx_n$ ⑤带入③ $x_n = (I + BR^{-1}B^TP_n)^{-1}Ax_{n-1}$

④→① ⑥→① $\lambda_{n-1} = 2[Q + A^TP_n(I + BR^{-1}B^TP_n)^{-1}A]x_{n-1} = 2P_{n-1}x_{n-1}$

递推式 $P_{k-1} = Q + A^TP_k(I + BR^{-1}B^TP_k)^{-1}A$ ⑨ Riccati 方程

$P_n = Q \Rightarrow P_{n-1} \Rightarrow P_{n-2} \Rightarrow \cdots$

$u_k = -\frac{1}{2}R^{-1}B^T\lambda_{k+1}$

$= -\frac{1}{2}R^{-1}B^T(2P_{k+1}x_{k+1}) = -R^{-1}B^TP_{k+1}(Ax_k + Bu_k)$

$u_k = -(R + B^TP_{k+1}B)^{-1}B^TP_{k+1}Ax_k$

考虑 $\sum_{k=0}^{\infty} x^TQx + u^TRu$ $n\to\infty$ $P_n = Q$ 要迭代无数次

一般⑨迭代几十次就会收敛 $P_k = P_{k-1} = P_{k-2} =$

P_{k+1} 实际上是 $P = Q + A^TP(I + BR^{-1}B^TP)^{-1}A$ 的解　Riccati 方程

P 取初值 Q　迭代 → P　　$u = -(R+B^TPB)^{-1}B^TPAx_k = -Kx_k$

$K = (R+B^TPB)^{-1}B^TPA$

$[k, s, E] = dlqr(A, B, Q, R)$　k　S=P　E 为特征值

$$P = Q + A^TPA - A^TPB(R+B^TPB)^{-1}B^TPA$$

矩阵求逆引理 $(A+BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}$

$P_{k+1} = Q + A^TP_k \underbrace{(I + BR^{-1}B^TP_k)^{-1}}_{4\times4\ \ 4\times1\ \ 1\times1\ \ 1\times4\ \ 4\times4}A$　$(4\times4)^{-1}$

$A = I$　$B = B$　$C = R^{-1}$　$D = B^TP_k$

$(I + BR^{-1}B^TP_k)^{-1} = I - B(R+B^TPB)^{-1}B^TP$

$P = Q + A^TP(I - B(R+B^TPB)^{-1}B^TP)A = \underbrace{Q + A^TPA - A^TPB(R+B^TPB)^{-1}B^TPA}_{1\times1\ \ 1\times4\ 4\times4\ 4\times1}$ ✓

u = δ　　B 4×1　　R 1×1　　P 4×4

总结　lqr

$\dot{e}_{rr} = Ae_{rr} + Bu$

① 离散化 $e_{rr}(k+1) = \bar{A}e_{rr}(k) + \bar{B}u(k)$

② 求解 Riccati 方程 $P = Q + \bar{A}^TP\bar{A} - \bar{A}^TP\bar{B}(R+\bar{B}^TP\bar{B})^{-1}\bar{B}^TP\bar{A}$

③ 求解　　$K = (R+\bar{B}^TP\bar{B})^{-1}\bar{B}^TP\bar{A}$

④ $u_k = -ke_{rr}(k)$